

Unifying Special Relativity and Quantum Mechanics

Lecture 6 — Massless Irreps Completed: the Wigner Phase, Parity, and the Vacuum;

From Particles to Fields: Creation and Annihilation Operators

(Typed version of the handwritten notes, with added explanations in blue boxes)

Added explanation (not in the original notes)

How to read this document. Same conventions as Lectures 1–5: everything in the normal text and in the yellow boxes is a faithful transcription of the handwritten notes (yellow boxes correspond to the results that were highlighted in the original, whether in yellow or in pink/red); everything inside a blue box like this one is extra material that I added. The figures are redrawn versions of the hand sketches. Short underlined phrases of the original are kept underlined; a few sentence-long underlines are rendered in italics instead, purely for typographic reasons (underlined text cannot break across lines).

Pagination of this scan. The scan continues the double-starred numbering sequence opened by the massless chapter of Lecture 5: its pages carry the circled numbers 7–19, i.e. they are [p. 7**]–[p. 19**], picking up exactly where [p. 6**] stopped. After [p. 19**] (and one blank page) the notes open yet another chapter, “From particles to fields”, and restart the page numbering from 1 a third time; to keep the three restarts apart, the markers of that part are written [p. 1***], . . . , [p. 8***].

Where we left off (recap of Lecture 5). For the massless case ($p^2 = 0$, $p^0 = |\vec{p}| > 0$) we chose the reference vector $p_{\text{ref}}^\mu = (k, 0, 0, k)$ with $k > 0$ arbitrary, found that the little group is $SE(2)$ with generators $T_x = J^1 + K^2$, $T_y = J^2 - K^1$, J^3 , discarded the continuous-spin irreps, and kept the *degenerate* one-dimensional irreps defined by

$$T_x|0, k\hat{z}, \lambda\rangle = T_y|0, k\hat{z}, \lambda\rangle = 0, \quad J^3|0, k\hat{z}, \lambda\rangle = \lambda|0, k\hat{z}, \lambda\rangle, \quad \lambda \in \tfrac{1}{2}\mathbb{Z},$$

with the covariant eigensystem P^μ , $P^2 = 0$, $-S_\hat{z}^\mu W_\mu/k$ (eigenvalue λ). What was still missing is the exact analogue of Steps 5–6 of the massive construction: generate the states with generic momentum, and work out how the full Poincaré group acts on them. That is where this lecture starts — and, once the massless case is closed, the notes pivot to the second part of the course: multi-particle states and, eventually, quantum fields.

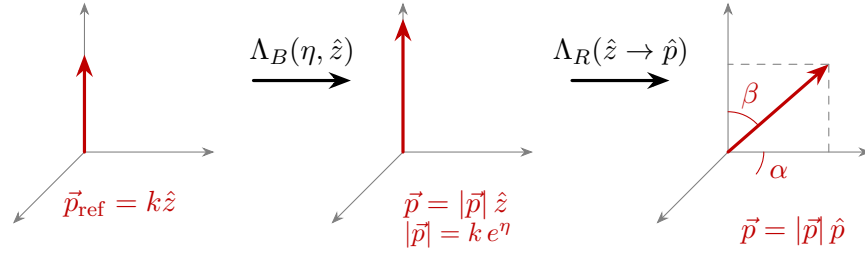
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1 Step 5: generation of states with generic momentum [p. 7**]

We start our considerations in momentum space.



1) We boost along $\hat{z}$.

$$\Lambda_B(\eta, \hat{z}) p_{\text{ref}} = \begin{pmatrix} \cosh \eta & 0 & 0 & \sinh \eta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \eta & 0 & 0 & \cosh \eta \end{pmatrix} \begin{pmatrix} k \\ 0 \\ 0 \\ k \end{pmatrix} = \begin{pmatrix} k(\cosh \eta + \sinh \eta) \\ 0 \\ 0 \\ k(\sinh \eta + \cosh \eta) \end{pmatrix}$$

and, using $\cosh \eta + \sinh \eta = e^\eta$,

$$= \begin{pmatrix} k e^\eta \\ 0 \\ 0 \\ k e^\eta \end{pmatrix} = \begin{pmatrix} |\vec{p}| \\ |\vec{p}| \hat{z} \end{pmatrix}, \quad \boxed{|\vec{p}| \equiv k e^\eta}$$

2) We rotate $\hat{z} \rightarrow \hat{p}$.

$$\Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(\eta, \hat{z}) p_{\text{ref}} = \begin{pmatrix} |\vec{p}| \\ |\vec{p}| \hat{p} \end{pmatrix}.$$

[p. 8**] We introduce the transformation

$$H(\hat{p}, |\vec{p}|/k) \equiv \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(\eta, \hat{z}), \quad H(\hat{p}, |\vec{p}|/k)^\mu{}_\nu p_{\text{ref}}^\nu = p^\mu$$

and the operator $U_H(\hat{p}, |\vec{p}|/k)$ that represents the action of this transformation on the states $|0, k\hat{z}, \lambda\rangle$.

To investigate the properties of $U_H|0, k\hat{z}, \lambda\rangle$, let us compute

$$P^\mu U_H|0, k\hat{z}, \lambda\rangle = U_H(U_H^{-1} P^\mu U_H)|0, k\hat{z}, \lambda\rangle = U_H H^\mu{}_\nu P^\nu|0, k\hat{z}, \lambda\rangle$$

$$= U_H H^\mu{}_\nu p_{\text{ref}}^\nu |0, k\hat{z}, \lambda\rangle = p^\mu \underline{U_H |0, k\hat{z}, \lambda\rangle}$$

$\rightsquigarrow$ still an eigenstate of P^μ , with eigenvalue $p^\mu = H^\mu{}_\nu p_{\text{ref}}^\nu$.
We define

$$|0, \vec{p}, \lambda\rangle = U_H(\hat{p}, |\vec{p}|/k) |0, k\hat{z}, \lambda\rangle$$

$\rightsquigarrow$ we immediately write the eigenvalue equations

$$P^\mu |0, \vec{p}, \lambda\rangle = p^\mu |0, \vec{p}, \lambda\rangle; \quad P^2 |0, \vec{p}, \lambda\rangle = 0.$$

What about the last one?

Added explanation (not in the original notes)

The massive/massless dictionary for Step 5. This is the word-for-word analogue of the massive standard boost of Lecture 4 ([p. 20*-22*]), $H(\hat{p}, |\vec{p}|/M) = \Lambda_R(\hat{z} \rightarrow \hat{p}) \Lambda_B(\eta, \hat{z})$ with $\sinh \eta = |\vec{p}|/M$; the only changes are that the reference vector is now light-like, and that the boost parameter is tied to the momentum through $e^\eta = |\vec{p}|/k$ instead of $\sinh \eta = |\vec{p}|/M$. The reason for the different relation is worth noticing: a boost cannot bring a light-like vector to rest — it can only *rescale* it, $(k, 0, 0, k) \mapsto e^\eta (k, 0, 0, k)$, sliding it along the light cone (the light cone of the Lecture 5 figure is the orbit, playing the role of the mass-shell hyperboloid). Also note that η ranges over all of $\mathbb{R}$: $\eta \rightarrow -\infty$ makes the momentum arbitrarily soft, $\eta \rightarrow +\infty$ arbitrarily hard, but never reaches $|\vec{p}| = 0$ — the tip of the cone ($p^\mu = 0$, “the vacuum”) is a separate orbit, treated at the end of this chapter. Finally, “the last one”: the rest-frame equation $-S_{\hat{z}}^\mu W_\mu/k$ (eigenvalue λ) of Lecture 5 is written in terms of the fixed vector $S_{\hat{z}}$, which is adapted to p_{ref} only — exactly as in the massive case ([p. 23*]) it must be transported along with the momentum before it can label the moving states. That is the next computation.

2 The helicity equation for generic momentum [p. 9**]

We elaborate the last equation as we did in the massive case. We use

$$S_p^\mu = H^\mu{}_\nu S_{\hat{z}}^\nu, \quad S_{\hat{z}}^\mu = (H^{-1})^\mu{}_\nu S_p^\nu$$

and evaluate

$$\begin{aligned} -\frac{S_p^\mu W_\mu}{k} |0, \vec{p}, \lambda\rangle &= -\frac{S_p^\mu W_\mu}{k} U_H |0, k\hat{z}, \lambda\rangle = -\frac{S_p^\mu}{k} U_H (U_H^{-1} W_\mu U_H) |0, k\hat{z}, \lambda\rangle \\ &= -\frac{S_p^\mu}{k} U_H H_\mu{}^\nu W_\nu |0, k\hat{z}, \lambda\rangle = U_H \frac{(-S_p^\mu)}{k} (H^{-1})^\mu{}_\nu W_\nu |0, k\hat{z}, \lambda\rangle \\ &= U_H \underbrace{\left(-\frac{S_{\hat{z}}^\mu W_\mu}{k} \right)} |0, k\hat{z}, \lambda\rangle = U_H \lambda |0, k\hat{z}, \lambda\rangle = \lambda |0, \vec{p}, \lambda\rangle. \end{aligned}$$

$$-\frac{S_p^\mu W_\mu}{k} |0, \vec{p}, \lambda\rangle = \lambda |0, \vec{p}, \lambda\rangle$$

◀ *what is this operator, physically?*

1) First, we write S_p^μ explicitly:

$$S_p^\mu = \begin{pmatrix} \sinh \eta \\ \cosh \eta \hat{p} \end{pmatrix}.$$

2) We then compute $-S_p^\mu W_\mu/k$:

$$\begin{aligned} -\frac{1}{k}(S_p^0 W_0 + S_p^i W_i) &= -\frac{1}{k}(S_p^0 W^0 - S_p^i W^i) \\ &= -\frac{1}{k} \left[\sinh \eta \vec{J} \cdot \vec{P} - \cosh \eta \hat{p} \cdot (\vec{J} P^0 + \vec{K} \times \vec{P}) \right]. \end{aligned}$$

[p. 10**] 3) We apply this operator to $|0, \vec{p}, \lambda\rangle$:

$$\begin{aligned} -\frac{S_p^\mu W_\mu}{k} |0, \vec{p}, \lambda\rangle &= -\frac{1}{k} \left(\sinh \eta \vec{J} \cdot \vec{p} - \cosh \eta \hat{p} \cdot \vec{J} |\vec{p}| \right) |0, \vec{p}, \lambda\rangle \\ &= -\frac{1}{k} (\sinh \eta - \cosh \eta) \vec{J} \cdot \vec{P} |0, \vec{p}, \lambda\rangle \\ &= \frac{e^{-\eta}}{k} \vec{J} \cdot \vec{P} |0, \vec{p}, \lambda\rangle = \frac{\vec{J} \cdot \vec{P}}{|\vec{p}|} |0, \vec{p}, \lambda\rangle = \vec{J} \cdot \hat{p} |0, \vec{p}, \lambda\rangle, \end{aligned}$$

using $e^{-\eta} = k/|\vec{p}| \rightsquigarrow$ HELICITY.

The key equations are

$$P^\mu |0, \vec{p}, \lambda\rangle = p^\mu |0, \vec{p}, \lambda\rangle, \quad P^2 |0, \vec{p}, \lambda\rangle = 0, \quad \vec{J} \cdot \hat{p} |0, \vec{p}, \lambda\rangle = \lambda |0, \vec{p}, \lambda\rangle.$$

Added explanation (not in the original notes)

Worked details of the cancellations. Two terms disappear silently in step 3): (i) $\hat{p} \cdot (\vec{K} \times \vec{P})$ acting on a momentum eigenstate gives $\hat{p} \cdot (\vec{K} \times \vec{p}) \propto \vec{p} \cdot (\vec{K} \times \vec{p}) = 0$ (a triple product with a repeated vector) — the same cancellation that made the boost part of $\vec{W}$ drop out of the massive helicity computation (Lecture 4, [p. 26*]); (ii) on the light cone $P^0 |0, \vec{p}, \lambda\rangle = |\vec{p}| |0, \vec{p}, \lambda\rangle$, which converts $\hat{p} \cdot \vec{J} P^0$ into $\vec{J} \cdot \vec{P}$ after multiplying by $\hat{p} \cdot \vec{p} = |\vec{p}|$ bookkeeping. Then $\sinh \eta - \cosh \eta = -e^{-\eta}$ turns the prefactor into $e^{-\eta}/k = 1/|\vec{p}|$: the arbitrary reference scale k drops out of the final answer, as it must — no physical statement can depend on the conventional choice of k (Lecture 5's remark that “its precise value is irrelevant”). The final operator $\vec{J} \cdot \hat{p}$ is exactly the helicity operator found for massive states in Lecture 4, but with one crucial difference developed on [p. 17**] below: with no rest frame available, λ is now the *only* spin-like label, and it will turn out to be Lorentz invariant.

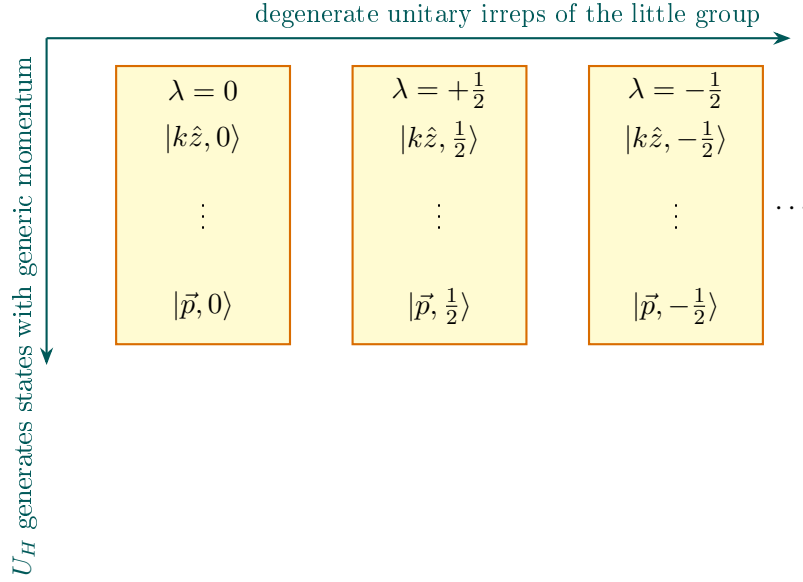
3 Step 6: massless unitary irreps of Poincaré [p. 10**]

The claim is that

$$\left\{ |0, \vec{p}, \lambda\rangle \right\}_{\text{with } \lambda \text{ fixed}}$$

furnish the representation space of a unitary irrep of the Poincaré group.

[p. 11**]



1) If we consider a generic Poincaré transformation $U(\Lambda, a)$ and act on a state $|\vec{p}, \lambda\rangle$, we should get a state with the same value of λ while generating all values of $\vec{p}$

$$\rightsquigarrow \left\{ |\vec{p}, \lambda\rangle \right\}_{\lambda \text{ fixed}} \quad \text{minimally invariant subspace under Poincaré.}$$

Thanks to $U(\Lambda, a) = U(a)U(\Lambda)$, we study separately translations and Lorentz.

Added explanation (not in the original notes)

Notation. From this page on the handwritten notes lighten the notation and drop the mass label, writing $|\vec{p}, \lambda\rangle$ for $|0, \vec{p}, \lambda\rangle$ (and $|k\hat{z}, \lambda\rangle$ for the reference state). Transcribed as written. The three towers in the figure are the massless counterpart of the table on [p. 27*] of Lecture 4 — with the crucial difference that each column now has *one single entry per momentum*: the little-group irreps are one-dimensional, so a massless “particle” carries no $(2s+1)$ -fold multiplet, just its helicity. States with $\lambda = +\frac{1}{2}$ and $\lambda = -\frac{1}{2}$ (or $+1$ and -1) live in *different* irreps — the physical weight of this fact is discussed on [p. 17**].

3.1 Action of space-time translations

This is the easy part. We write

$$U(a)|\vec{p}, \lambda\rangle = e^{ia_\mu P^\mu}|\vec{p}, \lambda\rangle = e^{ia \cdot p}|\vec{p}, \lambda\rangle \quad \text{multiplication by a phase factor.}$$

[p. 12**]

$$U(a)|\vec{p}, \lambda\rangle = e^{ip \cdot a}|\vec{p}, \lambda\rangle, \quad U(a)U(b) = U(a+b)$$

► the composition rule is verified; we stay within the same subspace.

3.2 Action of Lorentz transformations

We compute $U(\Lambda)|\vec{p}, \lambda\rangle$. First, we act on this state with P^μ :

$$P^\mu U(\Lambda)|\vec{p}, \lambda\rangle = U(\Lambda)(U(\Lambda)^{-1}P^\mu U(\Lambda))|\vec{p}, \lambda\rangle \stackrel{\text{“usual” computation}}{=} p_\Lambda^\mu U(\Lambda)|\vec{p}, \lambda\rangle$$

$\rightsquigarrow U(\Lambda)|\vec{p}, \lambda\rangle$ is eigenstate of P^μ with eigenvalue $p_\Lambda^\mu = \Lambda^\mu{}_\nu p^\nu$.

We already know how to write an eigenstate of P^μ with eigenvalue p_Λ (using the definition at p. 8**):

$$|\vec{p}_\Lambda, \lambda\rangle = U_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k)|k\hat{z}, \lambda\rangle.$$

What is the relation between $|\vec{p}_\Lambda, \lambda\rangle$ and $U(\Lambda)|\vec{p}, \lambda\rangle$? Let us define $U_{H_\Lambda} \equiv U_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k)$.

[p. 13**]

$$U(\Lambda)|\vec{p}, \lambda\rangle = U(\Lambda)U_H|k\hat{z}, \lambda\rangle = U_{H_\Lambda} \underbrace{(U_{H_\Lambda}^{-1}U(\Lambda)U_H)}_{\text{little group}}|k\hat{z}, \lambda\rangle \stackrel{(*)}{=}$$

In momentum space:

$$p_{\text{ref}} \xrightarrow{H} p = Hp_{\text{ref}} \xrightarrow{\Lambda} p_\Lambda = \Lambda p \xrightarrow{H_\Lambda^{-1}} p_{\text{ref}}$$

The combined transformation $H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k)^{-1}\Lambda H(\hat{p}, |\vec{p}|/k)$ leaves p_{ref} invariant, and must be an element of the little group:

$$H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k)^{-1}\Lambda H(\hat{p}, |\vec{p}|/k) \in SE(2)$$

$$U_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k)^{-1}U(\Lambda)U_H(\hat{p}, |\vec{p}|/k)$$

represents the action of a little-group element on the reference state $|k\hat{z}, \lambda\rangle$.

Which element of $SE(2)$? In full generality, we will have

$$U_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k)^{-1}U(\Lambda)U_H(\hat{p}, |\vec{p}|/k) = \exp(-i\alpha T_x - i\beta T_y) \exp(-i\theta J^3).$$

Since $T_x|k\hat{z}, \lambda\rangle = T_y|k\hat{z}, \lambda\rangle = 0$, we find [p. 14**]

$$\begin{aligned} U_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k)^{-1}U(\Lambda)U_H(\hat{p}, |\vec{p}|/k)|k\hat{z}, \lambda\rangle &= \exp(-i\alpha T_x - i\beta T_y) \exp(-i\theta J^3)|k\hat{z}, \lambda\rangle \\ &= \exp(-i\theta\lambda) \exp(-i\alpha T_x - i\beta T_y)|k\hat{z}, \lambda\rangle \\ &= e^{-i\theta\lambda}|k\hat{z}, \lambda\rangle. \end{aligned}$$

$$U_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k)^{-1}U(\Lambda)U_H(\hat{p}, |\vec{p}|/k)|k\hat{z}, \lambda\rangle = e^{-i\theta(\Lambda, p)\lambda}|k\hat{z}, \lambda\rangle$$

◀ $\theta(\Lambda, p)$ is called *Wigner phase*.

Going back to $\circledast$:

$$\stackrel{(*)}{=} U_H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k) e^{-i\theta(\Lambda, p)\lambda}|k\hat{z}, \lambda\rangle = e^{-i\theta(\Lambda, p)\lambda}|\vec{p}_\Lambda, \lambda\rangle.$$

We thus get

$$U(\Lambda)|\vec{p}, \lambda\rangle = e^{-i\theta(\Lambda, p)\lambda}|\vec{p}_\Lambda, \lambda\rangle$$

► if we apply a Lorentz transformation, we do change the momentum but we do not change the helicity $\rightsquigarrow$ we stay within the same subspace.

Added explanation (not in the original notes)

Why the little-group element can be written in that factorized form. The factorization $\exp(-i\alpha T_x - i\beta T_y) \exp(-i\theta J^3)$ is nothing but the $SE(2)$ factorization proved in Lecture 1 ([p. 8]): every Euclidean transformation splits uniquely as (translation)·(rotation), $E(R, \vec{a}) = T(\vec{a})R$, with the rotation acting *first*. Here T_x, T_y are the “translations” and J^3 the “rotation” of the little-group $SE(2)$ (Lecture 5). The order then does all the work: J^3 hits the reference state first and produces the phase $e^{-i\theta\lambda}$; the translation factor is then harmless because $T_{x,y}|k\hat{z}, \lambda\rangle = 0$ implies, by expanding the exponential series, $\exp(-i\alpha T_x - i\beta T_y)|k\hat{z}, \lambda\rangle = |k\hat{z}, \lambda\rangle$ — every term of the series beyond $\mathbb{1}$ annihilates the state. (This is exactly why the degenerate irreps of Lecture 5 were selected: on them, the non-compact part of $SE(2)$ acts trivially and only the compact $SO(2)$ survives, as a phase.)

Wigner rotation vs. Wigner phase. Comparing with the massive formula of Lecture 4,

$$\begin{aligned} \text{massive: } U(\Lambda)|M, \vec{p}, s, \lambda\rangle &= \sum_{\lambda'} [D_s(W(\Lambda, p))]_{\lambda'\lambda} |M, \vec{p}_\Lambda, s, \lambda'\rangle \\ \text{massless: } U(\Lambda)|\vec{p}, \lambda\rangle &= e^{-i\theta(\Lambda, p)\lambda} |\vec{p}_\Lambda, \lambda\rangle, \end{aligned}$$

the $(2s+1) \times (2s+1)$ Wigner-rotation matrix has collapsed to a 1×1 unitary matrix — a phase — because the little-group irrep is one-dimensional. There is no sum over λ' : helicity cannot be rotated away. The same trilogy of structures has now appeared three times, in strictly increasing complexity: for $SE(2)$ itself the sandwich $U^{-1}UU$ was the *identity* (Lecture 2, trivial little group); for massive Poincaré it is a *rotation matrix* acting on the spin index; for massless Poincaré it is a *phase* multiplying the state.

4 Composition rule and unitarity [p. 15**]

2) The operators $U(a)$, $U(\Lambda)$ and $U(\Lambda, a)$ verify the composition rule of Poincaré $\rightsquigarrow$ they provide an homomorphism. We checked already that $U(a)U(b) = U(a+b)$. Similarly, it is possible to check that $U(\Lambda_2)U(\Lambda_1) = U(\Lambda_2\Lambda_1)$.

3) The representation is unitary. We introduce, in each subspace with λ fixed,

$$\begin{aligned} \langle \vec{p}, \lambda | \vec{q}, \lambda \rangle &= (2\pi)^3 2E_q \delta(\vec{p} - \vec{q}), \quad E_q = |\vec{q}|, \\ \mathbb{1} &= \int \frac{d^3\vec{p}}{(2\pi)^3 2E_p} |\vec{p}, \lambda\rangle \langle \vec{p}, \lambda|. \end{aligned}$$

Unitarity follows from the fact that we essentially reduced the action of Lorentz transformations to a phase factor.

Comment. The Wigner phase enjoys the property:

If Λ is a pure rotation around $\hat{p}$ with some angle ϕ , $\Lambda = \Lambda_R(\hat{p}, \phi)$:

$$\theta(\Lambda_R(\hat{p}, \phi), p) = \phi.$$

Added explanation (not in the original notes)

Filling in the unitarity argument. The proof is the same change-of-variables argument used twice already (Lecture 2, [p. 26], and Lecture 5 for the massive Poincaré case): insert the completeness relation into $U(\Lambda, a)U(\Lambda, a)^\dagger$; the transformation law contributes only the phases $e^{\pm i(a \cdot p_\Lambda + \theta(\Lambda, p) \lambda)}$, which cancel between ket and bra, leaving $\int \frac{d^3 \vec{p}}{(2\pi)^3 2E_p} |\vec{p}_\Lambda, \lambda\rangle \langle \vec{p}_\Lambda, \lambda|$; finally $d^3 \vec{p}/(2E_p)$ is the Lorentz-invariant measure on the light cone (the $M \rightarrow 0$ limit of the mass-shell measure justified in Lecture 5 via $d^4 p \delta(p^2) \theta(p^0)$), so the integral over $\vec{p}$ can be traded for the integral over $\vec{p}_\Lambda$ and completeness reassembles into $\mathbb{1}$. Note that with the normalization $E_q = |\vec{q}|$ the massless states are “relativistically normalized” exactly like the massive ones — this uniform convention is what will let the creation-operator formalism of the next chapter treat both cases at once.

[p. 16**] $\hookrightarrow$ Flavour of the proof. From the definition, in momentum space,

$$H(\hat{p}_\Lambda, |\vec{p}_\Lambda|/k)^{-1} \Lambda H(\hat{p}, |\vec{p}|/k) \in SE(2).$$

We take $\Lambda \equiv \Lambda_R(\hat{p}, \phi)$; consequently

$$\Lambda_R(\hat{p}, \phi) p = p \Rightarrow p_\Lambda = p,$$

$$H(\hat{p}, |\vec{p}|/k)^{-1} \Lambda_R(\hat{p}, \phi) H(\hat{p}, |\vec{p}|/k) \in SE(2).$$

Expanding H ,

$$\Lambda_B(\eta, \hat{z})^{-1} \underbrace{\Lambda_R(\hat{z} \rightarrow \hat{p})^{-1} \Lambda_R(\hat{p}, \phi) \Lambda_R(\hat{z} \rightarrow \hat{p})}_{\text{}} \Lambda_B(\eta, \hat{z}) \stackrel{(*)}{=}$$

In this sequence of transformations:

$\Lambda_R(\hat{z} \rightarrow \hat{p})$ brings $\hat{z}$ into $\hat{p}$;

$\Lambda_R(\hat{p}, \phi)$ performs a rotation by ϕ around $\hat{p}$;

$\Lambda_R(\hat{z} \rightarrow \hat{p})^{-1}$ brings $\hat{p}$ back to $\hat{z}$.

The overall transformation corresponds to a rotation by an angle ϕ around $\hat{z}$:

$$\stackrel{(*)}{=} \Lambda_B(\eta, \hat{z})^{-1} \Lambda_R(\hat{z}, \phi) \Lambda_B(\eta, \hat{z}) = \Lambda_R(\hat{z}, \phi) = e^{-i\phi J_{4\text{vec}}^3}$$

which implies

$$\theta(\Lambda_R(\hat{p}, \phi), p) = \phi$$

Added explanation (not in the original notes)

The two lemmas used in the last line. (i) The conjugation $\Lambda_R(\hat{z} \rightarrow \hat{p})^{-1} \Lambda_R(\hat{p}, \phi) \Lambda_R(\hat{z} \rightarrow \hat{p}) = \Lambda_R(\hat{z}, \phi)$ is the statement that a rotation about a rotated axis is the rotated rotation — the group-level version of “ $\vec{J}$ transforms as a vector under rotations”, $[J^i, J^j] = i\varepsilon_{ijk} J^k$, and the same manipulation used in Lecture 5’s proof that $W_B(\Lambda_R, p) = \Lambda_R$ for the massive spin-polarization basis ([p. 42*–45*]). (ii) The final cancellation $\Lambda_B(\eta, \hat{z})^{-1} \Lambda_R(\hat{z}, \phi) \Lambda_B(\eta, \hat{z}) = \Lambda_R(\hat{z}, \phi)$ holds because a rotation *about* $\hat{z}$ commutes with a boost *along* $\hat{z}$: at the algebra level $[J^3, K^3] = 0$ (from $[J^i, K^j] = i\varepsilon_{ijk} K^k$ with $i = j = 3$) — this was precisely the last step of that Lecture 5 proof as well.

Why this particular property matters physically. It confirms the interpretation of λ : under a rotation by ϕ about the direction of motion, the state picks up $e^{-i\phi\lambda}$ — the defining behaviour of a state with angular momentum λ along that axis. It also exposes the by-now-familiar 2π subtlety: for half-integer λ , a 2π rotation gives $e^{-2\pi i\lambda} = -1$. As discussed in Lecture 4 ([p. 14*–15*]), this sign is exactly the projective freedom allowed by quantum mechanics, and it is what admits $\lambda \in \frac{1}{2}\mathbb{Z}$ rather than only integers.

5 Physical considerations [p. 17**]

- 1) We associate to each unitary irrep of Poincaré a massless relativistic particle.
- 2) For a massless particle, the helicity is a Lorentz invariant quantity.
- 3) At the most primitive level, *massless particles are defined by one degree of freedom: their helicity λ*

$\hookrightarrow$ photons with helicity $+1$ are distinct from photons with helicity -1 .

Comment. However, indeed we are used to think about photons as particles with two transverse polarizations, helicities ± 1 . Why?

In QED, there is an additional space-time symmetry in the structure of the theory, parity.

► *Parity is not continuously connected to the identity, and its action on the physical state is not captured by our $U(\Lambda)$.*

[p. 18**] ► What does parity do on the states $|\vec{p}, \lambda\rangle$?

Parity changes $\vec{p} \rightarrow -\vec{p}$ but not the spin. Consequently, and crucially, it changes $\lambda \rightarrow -\lambda$.

The bottom line is that, if we include parity as a fundamental symmetry, the states $\{|\vec{p}, \lambda\rangle\}_{\lambda \text{ fixed}}$ do not form anymore an irreducible representation

$\rightsquigarrow$ if we want to implement parity as a symmetry, we are forced to “pair up” states with opposite helicities

$$\left\{ |\vec{p}, \pm\lambda\rangle \right\}_{\lambda \text{ fixed}}$$

Added explanation (not in the original notes)

Why helicity is Lorentz invariant only in the massless case. For a massive particle, helicity is frame-dependent: an observer moving faster than the particle sees $\vec{p}$ reversed but the spin unchanged, so λ flips — “overtaking” is always possible since the particle moves at $v < c$ (this was anticipated in the Lecture 4 blue box on [p. 26*]). A massless particle moves at exactly c and can never be overtaken: no orthochronous Lorentz transformation reverses its momentum direction relative to its spin, and indeed the boxed transformation law $U(\Lambda)|\vec{p}, \lambda\rangle = e^{-i\theta\lambda}|\vec{p}_\Lambda, \lambda\rangle$ contains no mechanism to change λ at all.

Why parity flips λ . Under parity, $\vec{p}$ is a polar vector ($\vec{p} \rightarrow -\vec{p}$) while $\vec{J}$ is an axial vector ($\vec{J} \rightarrow +\vec{J}$): the product $\vec{J} \cdot \vec{p} \rightarrow -\vec{J} \cdot \vec{p}$, so a λ eigenstate is mapped to a $-\lambda$ eigenstate. Physics dictionary: electromagnetism and gravity conserve parity, so the “photon” is the parity-paired doublet $\lambda = \pm 1$ (the two transverse polarizations of light — there is no $\lambda = 0$ state: that would be a longitudinal photon, absent precisely because the little-group irrep is one-dimensional), and the graviton is the pair $\lambda = \pm 2$. The weak interactions, by contrast, violate parity maximally: in the Standard Model with massless neutrinos, the neutrino appears only with $\lambda = -\frac{1}{2}$ (and the antineutrino with $+\frac{1}{2}$) — a single, unpaired Wigner irrep. Nothing in Wigner’s classification forces the pairing; it is an *additional* discrete symmetry requirement, imposed (or not) by the

dynamics.

The vacuum state

We now consider the case

$$p^2 = 0 \quad \text{with} \quad p^0 = |\vec{p}| = 0.$$

We take $p_{\text{ref}}^\mu = (0, 0, 0, 0)$ and

$$P^\mu |0\rangle = 0 \rightsquigarrow U(a)|0\rangle = e^{ia_\mu P^\mu} |0\rangle = |0\rangle$$

→ the vacuum is invariant under space-time translations. What about the other transformations?
[p. 19**] By definition, *we define the vacuum as the state that does not transform under the entire Poincaré group*:

$$J^{\mu\nu} |0\rangle = 0, \quad U(\Lambda) |0\rangle = \exp\left(-\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}\right) |0\rangle = |0\rangle.$$

Added explanation (not in the original notes)

Closing the classification. This settles the last case of the momentum-orbit classification of Lecture 3 ([p. 2*]): the “null case” $p^\mu = 0$. Its orbit is a single point, invariant under the whole Lorentz group, so the little group is all of $SO^\uparrow(1, 3)$ and the natural (and physically chosen) representation is the trivial one-dimensional one: a unique state $|0\rangle$ on which every $U(\Lambda, a)$ acts as the identity — “the empty state looks the same to every inertial observer”, as anticipated in the Lecture 3 blue box. Note the logical shift in the last yellow box: translation invariance *followed* from $P^\mu |0\rangle = 0$, while Lorentz invariance is *imposed as a definition* ($J^{\mu\nu} |0\rangle = 0$) — it selects, among the states with $p^\mu = 0$, the one carrying the trivial irrep. The vacuum’s Poincaré invariance, innocuous-looking here, becomes a load-bearing assumption in the next chapter: $H_0 |0\rangle = 0$, $U(\Lambda, a) |0\rangle = |0\rangle$ are used explicitly in the computations on [p. 6***] and [p. 7***].

6 From particles to fields [p. 1***]

We constructed states describing relativistic QM particles, on which Poincaré transformations act as unitary operators.

► Good news: this picture is in agreement with Wigner’s theorem [according to which in QM symmetry transformations are implemented by either unitary (and linear) operators or by anti-unitary (and anti-linear) operators].

► Bad news: the states $|M, \vec{p}, s, \sigma\rangle$ have fixed momentum and, as such, they are *completely delocalized in space*.

[This suggests that these states are not the appropriate objects to encode relativistic causality.]

One might try to switch to position eigenstates, that is, to completely localize a particle in space.

[p. 2***] [This also leads to difficulties: attempting to localize a particle arbitrarily well requires large energies, and at sufficiently small distances pair-production of particles becomes possible → a description in terms of single-particle states alone is not adequate.]

We are, therefore, led to a modification of the framework:

- 1) First, we must allow for states with an arbitrary number of particles.
- 2) Second, instead of localizing states, we localize observations.

Added explanation (not in the original notes)

The pivot of the second half of the course. Wigner’s theorem is the statement that already motivated the search for *unitary* irreps back in Lecture 1 ([p. 16]-blue box): symmetries must preserve transition probabilities $|\langle\phi|\psi\rangle|^2$, which forces (anti)unitarity. The “good news” is that the construction of Lectures 3–6 realizes this program completely.

The “bad news” deserves unpacking, because it is the conceptual birth of quantum field theory. A momentum eigenstate is a plane wave spread over all of space, so it cannot even formulate a question like “can a measurement *here* influence a measurement *there*, outside the light cone?”. The obvious fix — Fourier-transform to position eigenstates — fails for two deep reasons: (i) kinematically, there is no fully satisfactory relativistic position operator (localizing a particle in a region smaller than its Compton wavelength $\sim 1/m$ requires momentum components with energies above the pair-production threshold $2m$); (ii) dynamically, once pair production is possible, particle number is not conserved, so a fixed- N Hilbert space cannot host the physics. The two numbered modifications answer the two problems in order: enlarge the state space to arbitrary particle number (Fock space, built next via creation operators), and attach locality to the *observables* — operators $\phi(x)$ labeled by spacetime points — rather than to the states. Causality then becomes a condition on (anti)commutators of observables at spacelike separation, which is exactly where the notes are heading ([p. 8***]).

7 Annihilation and creation operators (Point 1) [p. 2***]

Introducing annihilation and creation operators is a very convenient way of dealing with multi-particle states.

[p. 3***] We define

$$|m, \vec{p}, \lambda\rangle = \sqrt{2E_p} a^\dagger(m, \vec{p}, \lambda)|0\rangle, \quad a(m, \vec{p}, \lambda)|0\rangle = 0, \quad E_p = (|\vec{p}|^2 + m^2)^{1/2}.$$

► $|m, \vec{p}, \lambda\rangle$ are precisely the single-particle states that belong to a given irrep of Poincaré. The index “ m ” represents the set of Casimir eigenvalues that defines the irrep.

Multiparticle states are defined by the subsequent application of creation operators:

$$|m_1, \vec{p}_1, \lambda_1; \dots; m_N, \vec{p}_N, \lambda_N\rangle = \prod_{k=1}^N \sqrt{2E_k} a^\dagger(m_k, \vec{p}_k, \lambda_k)|0\rangle, \quad E_k = (|\vec{p}_k|^2 + m_k^2)^{1/2} \forall k.$$

These operators satisfy the algebra

$$\begin{aligned}
[a(m, \vec{p}, \lambda), a^\dagger(m', \vec{q}, \lambda')] &= (2\pi)^3 \delta(\vec{p} - \vec{q}) \delta_{\lambda\lambda'} \delta_{mm'} && \text{INTEGER SPIN} \\
\{a(m, \vec{p}, \lambda), a^\dagger(m', \vec{q}, \lambda')\} &= (2\pi)^3 \delta(\vec{p} - \vec{q}) \delta_{\lambda\lambda'} \delta_{mm'} && \text{HALF-INTEGER SPIN} \\
[a, a] &= [a^\dagger, a^\dagger] = 0; && \{a, a\} = \{a^\dagger, a^\dagger\} = 0
\end{aligned}$$

[p. 4***] with the condition

$$\langle m, \vec{p}, \lambda | m', \vec{q}, \lambda' \rangle = (2\pi)^3 (2E_p) \delta_{mm'} \delta_{\lambda\lambda'} \delta(\vec{p} - \vec{q}).$$

Comments.

1) A description in terms of creation and annihilation operators is valid in the case of free particles

$\hookrightarrow$ in $|m_1, \vec{p}_1, \lambda_1; \dots; m_N, \vec{p}_N, \lambda_N\rangle$ we “add particles without disturbing the others”,

and each particle obeys its own on-shell condition $E_i = (|\vec{p}_i|^2 + m_i^2)^{1/2}$.

2) In a free theory, we can immediately write the Hamiltonian and momentum operators

$$\begin{aligned}
H_0 &= \sum_m \left[\sum_\lambda \int \frac{d^3 \vec{p}}{(2\pi)^3} E_p a^\dagger(m, \vec{p}, \lambda) a(m, \vec{p}, \lambda) \right] \\
\vec{P} &= \sum_m \left[\sum_\lambda \int \frac{d^3 \vec{p}}{(2\pi)^3} \vec{p} a^\dagger(m, \vec{p}, \lambda) a(m, \vec{p}, \lambda) \right]
\end{aligned}$$

Added explanation (not in the original notes)

Consistency of the definitions. The normalization $|m, \vec{p}, \lambda\rangle = \sqrt{2E_p} a^\dagger|0\rangle$ together with the algebra reproduces exactly the relativistic inner product introduced for the one-particle irreps (Lecture 5 for the massive case, [p. 15**] above for the massless one): $\langle m, \vec{p}, \lambda | m', \vec{q}, \lambda' \rangle = \sqrt{2E_p} \sqrt{2E_q} \langle 0 | a a^\dagger | 0 \rangle = \sqrt{2E_p} \sqrt{2E_q} (2\pi)^3 \delta(\vec{p} - \vec{q}) \delta_{mm'} \delta_{\lambda\lambda'}$, and the delta function sets $E_q = E_p$ — the boxed condition on [p. 4***]. (For either statistics: the $\pm$ term $\langle 0 | a^\dagger a | 0 \rangle$ vanishes by $a|0\rangle = 0$.)

The spin–statistics pairing is an input here. The notes *assign* commutators to integer spin and anticommutators to half-integer spin. At this stage of the course this is an empirical rule (it encodes Bose–Einstein vs. Fermi–Dirac statistics; note that the anticommutator $\{a^\dagger, a^\dagger\} = 0$ gives $a^\dagger(m, \vec{p}, \lambda)^2 = 0$, i.e. the Pauli exclusion principle, for free). The *spin–statistics theorem* — the statement that this pairing is forced by Lorentz invariance + causality + positivity of the energy — is one of the payoffs of the field-theoretic framework being set up, and presumably comes later in the course. The remarkable point emphasized on [p. 5***] is that everything in *this* chapter works identically for both choices.

Normal ordering, quietly. In H_0 every $a^\dagger$ stands to the left of every a ; together with $a|0\rangle = 0$ this makes $H_0|0\rangle = 0$ automatic — the vacuum has zero energy by construction, consistent with its Poincaré invariance ([p. 19**]). Starting instead from a classical expression like $\frac{1}{2} \sum (aa^\dagger + a^\dagger a)$ would produce an (infinite) constant; the choice made here silently fixes that ambiguity.

[p. 5***] $\hookrightarrow$ Flavour of the proof. Interestingly, the result is valid whatever the statistics of annihilation and creation operators. Let us introduce the following notation:

$$[A, B] = AB - BA \equiv [A, B]_-, \quad \{A, B\} = AB + BA \equiv [A, B]_+,$$

such that we have the following identity:

$$[AB, C] = A[B, C]_{\pm} \mp [A, C]_{\pm} B \quad (\star)$$

Consequently,

$$\begin{aligned} [H_0, a^\dagger(m, \vec{p}, \lambda)] &= \sum_{m'} \sum_{\lambda'} \int \frac{d^3 \vec{k}}{(2\pi)^3} E_k \underbrace{[a^\dagger(m', \vec{k}, \lambda') a(m', \vec{k}, \lambda'), a^\dagger(m, \vec{p}, \lambda)]}_{\text{we use } (\star)} \\ &= \sum_{m'} \sum_{\lambda'} \int \frac{d^3 \vec{k}}{(2\pi)^3} E_k a^\dagger(m', \vec{k}, \lambda') [a(m', \vec{k}, \lambda'), a^\dagger(m, \vec{p}, \lambda)]_{\pm} \\ &= \sum_{m'} \sum_{\lambda'} \int \frac{d^3 \vec{k}}{(2\pi)^3} E_k a^\dagger(m', \vec{k}, \lambda') (2\pi)^3 \delta_{mm'} \delta_{\lambda\lambda'} \delta(\vec{k} - \vec{p}) \\ &= E_p a^\dagger(m, \vec{p}, \lambda). \end{aligned}$$

[p. 6***]

$$[H_0, a^\dagger(m, \vec{p}, \lambda)] = E_p a^\dagger(m, \vec{p}, \lambda) \quad \underline{\text{independent on the statistics}}$$

This computation generalizes to

$$[H_0, a^\dagger(m_1, \vec{p}_1, \lambda_1) \cdots a^\dagger(m_N, \vec{p}_N, \lambda_N)] = (E_{p_1} + \cdots + E_{p_N}) a^\dagger(m_1, \vec{p}_1, \lambda_1) \cdots a^\dagger(m_N, \vec{p}_N, \lambda_N)$$

and we finally get (using $H_0|0\rangle = 0$)

$$H_0 |m_1, \vec{p}_1, \lambda_1; \dots; m_N, \vec{p}_N, \lambda_N\rangle = (E_{p_1} + \cdots + E_{p_N}) |m_1, \vec{p}_1, \lambda_1; \dots; m_N, \vec{p}_N, \lambda_N\rangle$$

$\hookrightarrow H_0$ counts the energies of each particle.

Added explanation (not in the original notes)

Checking the identity $(\star)$ and the steps of the proof. Expand both versions: with commutators (lower signs), $A[B, C]_- + [A, C]_- B = ABC - ACB + ACB - CAB = [AB, C]$; with anticommutators (upper signs), $A[B, C]_+ - [A, C]_+ B = ABC + ACB - ACB - CAB = [AB, C]$. ✓ The identity converts the *commutator* of H_0 (which must be a commutator: H_0 is an observable) into the *elementary brackets* of the algebra — commutators for bosons, anticommutators for fermions — which is why the final answer is statistics-blind. In the second line, the term $\mp[a^\dagger(m', \vec{k}, \lambda'), a^\dagger(m, \vec{p}, \lambda)]_{\pm} a(m', \vec{k}, \lambda')$ was dropped because $[a^\dagger, a^\dagger]_{\pm} = 0$ in both statistics (last line of the [p. 3***] box). The generalization to N creation operators follows by iterating the Leibniz-like rule $[H_0, XY] = [H_0, X]Y + X[H_0, Y]$ (valid for genuine commutators), each factor contributing its own E_{p_i} ; the analogous computation with $\vec{P}$ gives total momentum $\vec{p}_1 + \cdots + \vec{p}_N$. Physical summary: $a^\dagger(m, \vec{p}, \lambda)$ adds energy E_p and momentum $\vec{p}$ to any state — free particles

superpose without interacting, each on its own mass shell (Comment 1 above).

8 Transformation properties under Poincaré [p. 6***]

We consider the massive case, and work in the spin-polarization basis:

$$|m, \vec{p}, \sigma\rangle = \sqrt{2E_p} a^\dagger(m, \vec{p}, \sigma)|0\rangle.$$

[p. 7***] Consider $|M, \vec{p}, s, \sigma\rangle = \sqrt{2E_p} a^\dagger(M, \vec{p}, s, \sigma)|0\rangle$ and we apply $U(\Lambda, a)$:

$$U(\Lambda, a)|M, \vec{p}, s, \sigma\rangle = (2E_p)^{1/2} U(\Lambda, a) a^\dagger(M, \vec{p}, s, \sigma)|0\rangle.$$

On the left-hand side, *we know what is this object*; on the right-hand side, we insert $U^{-1}U = \mathbb{1}$ and use $U(\Lambda, a)|0\rangle = |0\rangle$:

$$e^{ip_\Lambda \cdot a} \sum_{\sigma'} [D_s(W_L(\Lambda, p))]_{\sigma'\sigma} |M, \vec{p}_\Lambda, s, \sigma'\rangle = (2E_p)^{1/2} U(\Lambda, a) a^\dagger(M, \vec{p}, s, \sigma) U(\Lambda, a)^{-1} |0\rangle.$$

We use $|M, \vec{p}_\Lambda, s, \sigma'\rangle = (2E_{p_\Lambda})^{1/2} a^\dagger(M, \vec{p}_\Lambda, s, \sigma')|0\rangle$:

$$\begin{aligned} e^{ip_\Lambda \cdot a} \sum_{\sigma'} [D_s(W_L(\Lambda, p))]_{\sigma'\sigma} (2E_{p_\Lambda})^{1/2} a^\dagger(M, \vec{p}_\Lambda, s, \sigma')|0\rangle \\ = (2E_p)^{1/2} U(\Lambda, a) a^\dagger(M, \vec{p}, s, \sigma) U(\Lambda, a)^{-1} |0\rangle. \end{aligned}$$

We get

$$U(\Lambda, a) a^\dagger(M, \vec{p}, s, \sigma) U(\Lambda, a)^{-1} = e^{ia \cdot p_\Lambda} \left(\frac{E_{p_\Lambda}}{E_p} \right)^{1/2} \sum_{\sigma'} [D_s(W_L(\Lambda, p))]_{\sigma'\sigma} a^\dagger(M, \vec{p}_\Lambda, s, \sigma')$$

By taking the Hermitian conjugate, [p. 8***]

$$U(\Lambda, a) a(M, \vec{p}, s, \sigma) U(\Lambda, a)^{-1} = e^{-ia \cdot p_\Lambda} \left(\frac{E_{p_\Lambda}}{E_p} \right)^{1/2} \sum_{\sigma'} [D_s(W_L(\Lambda, p))]_{\sigma'\sigma}^* a(M, \vec{p}_\Lambda, s, \sigma')$$

We now consider the massless case: the spirit of the computation is the same, with the key difference that

$$(D_s(W_L(\Lambda, p)))_{\sigma'\sigma} \longrightarrow e^{-i\theta(\Lambda, p) \cdot \lambda} \delta_{\lambda'\lambda}.$$

$$U(\Lambda, a) a^\dagger(\vec{p}, \lambda) U(\Lambda, a)^{-1} = e^{ia \cdot p_\Lambda} \left(\frac{E_{p_\Lambda}}{E_p} \right)^{1/2} e^{-i\theta(\Lambda, p) \cdot \lambda} a^\dagger(\vec{p}_\Lambda, \lambda)$$

$$U(\Lambda, a) a(\vec{p}, \lambda) U(\Lambda, a)^{-1} = e^{-ia \cdot p_\Lambda} \left(\frac{E_{p_\Lambda}}{E_p} \right)^{1/2} e^{+i\theta(\Lambda, p) \cdot \lambda} a(\vec{p}_\Lambda, \lambda)$$

Added explanation (not in the original notes)

Anatomy of the derivation. The logic is a matching argument: the same state, $U(\Lambda, a)|M, \vec{p}, s, \sigma\rangle$, is evaluated once with the *known* one-particle transformation law (Lecture 5, in the spin-polarization basis) and once by conjugating the creation operator with $U(\Lambda, a)$; since $|0\rangle$ is Poincaré invariant ([p. 19**] — this is where that “definition” earns its keep), the two right-hand sides act on the same vacuum, and comparing them determines the conjugated operator. The energy ratio $(E_{p_\Lambda}/E_p)^{1/2}$ is pure bookkeeping between the $\sqrt{2E}$ normalization of the states and the δ -function normalization of the operators: it is exactly what makes the transformation law consistent with the Lorentz-invariant measure $d^3\vec{p}/(2E_p)$.

A small notational remark. The notes here write $W_L(\Lambda, p)$ for the Wigner rotation; the subscript L refers to the standard transformation of the spin-polarization basis, the *direct boost* $L(p) = \Lambda_B(|\vec{p}|/M, \hat{p})$ in Weinberg’s notation — i.e. the same object called $W_B(\Lambda, p) = L(p_\Lambda)^{-1}\Lambda L(p)$ in Lecture 5 ([p. 40*–41*]).

Why a carries D^* (and the massless phase flips sign). Taking the Hermitian conjugate of the $a^\dagger$ law reverses the order of the U ’s, conjugates the phase, and complex-conjugates the matrix elements; renaming $U(\Lambda, a) \rightarrow U(\Lambda, a)$ back via $U^\dagger = U^{-1}$ (unitarity!) gives the boxed result. In representation-theory language: annihilation operators transform in the *complex-conjugate* representation D_s^* . For the massless case D_s is the 1×1 phase $e^{-i\theta\lambda}$, whose conjugate is $e^{+i\theta\lambda}$ — hence the sign flip between the two boxes.

Where this is going. These two boxes are the raw material for “Point 2” of [p. 2***]: local observables. Neither $a^\dagger(\vec{p}, \lambda)$ nor $a(\vec{p}, \lambda)$ is attached to a spacetime point — the next step (opened by the final lines below) is to build linear combinations $\phi(x) \sim \int \frac{d^3\vec{p}}{(2\pi)^3 2E_p} (e^{-ip \cdot x} a + e^{+ip \cdot x} a^\dagger)$ engineered so that $U(\Lambda, a)\phi(x)U(\Lambda, a)^{-1}$ acts geometrically on the label x — a *quantum field*. The transformation laws just derived, with their D_s matrices and energy ratios, are precisely the constraints that fix the coefficient functions in front of a and $a^\dagger$.

Quantum fields of any spin (Point 2) [p. 8***]

In NRQM, any Hermitian operator is an observable. This cannot be true if we include special relativity. In QM, we implement relativistic causality as follows...

(End of Lecture 6 — the notes continue with the implementation of relativistic causality and the construction of quantum fields.)